

Correlation Analysis of Chronic Disease Indicators and Health Behaviors in the United States

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Chronic diseases such as diabetes, heart disease, and obesity represent major public health challenges. Many of these conditions are influenced by modifiable behaviors, including physical activity, smoking, and diet.

Understanding the relationships between chronic disease indicators and health behaviors can give us prevention strategies.

In this project we aproach the following problems:

How do chronic diseases vary across the US?

How are unhealthy behaviors, particularly binge drinking, related to mortality rates from chronic liver disease?

Methods

DATA COLECTION

U.S. Chronic Disease Indicators (CDC 2019-2022)

DATA PREPROCESSING

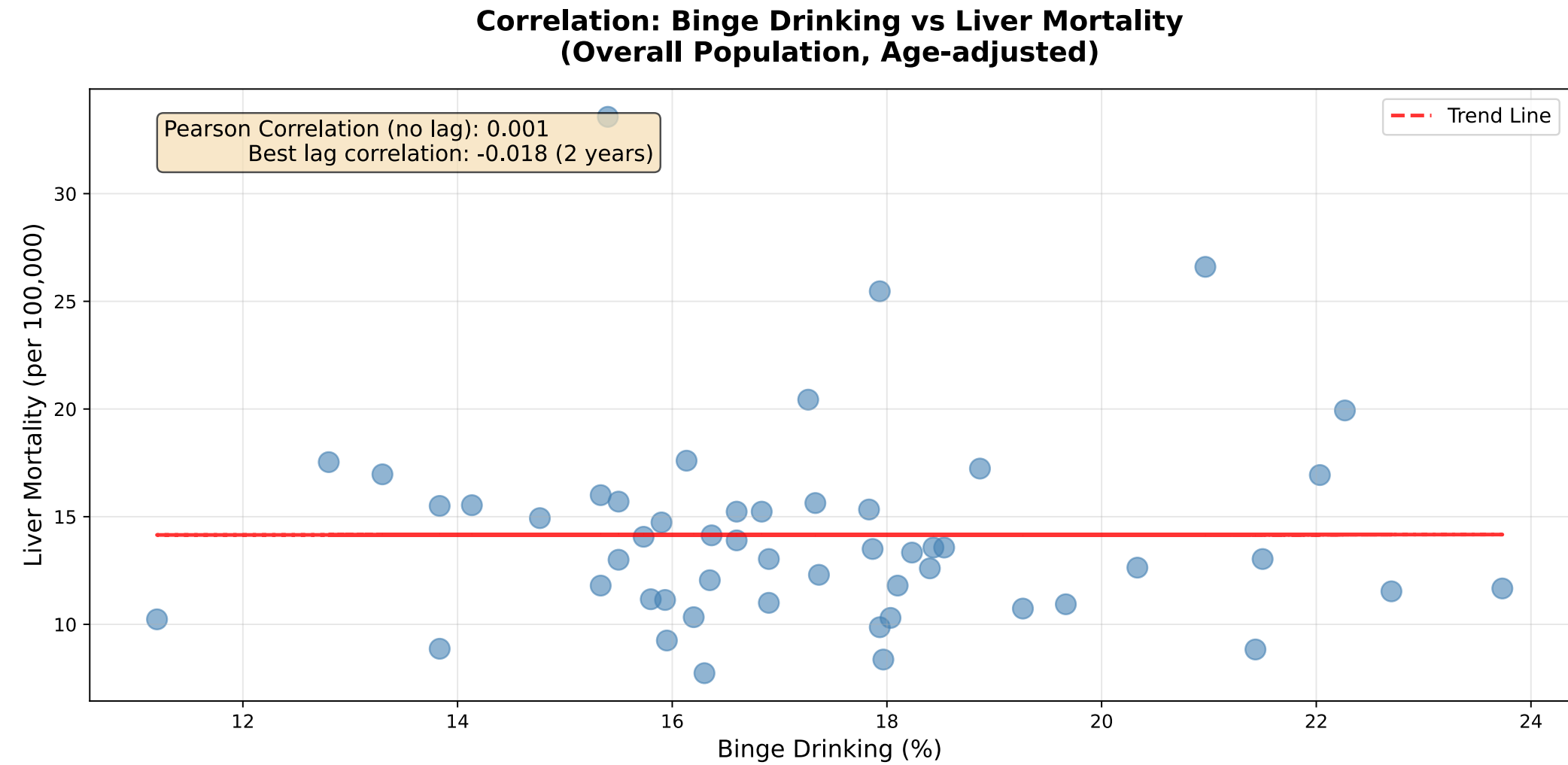
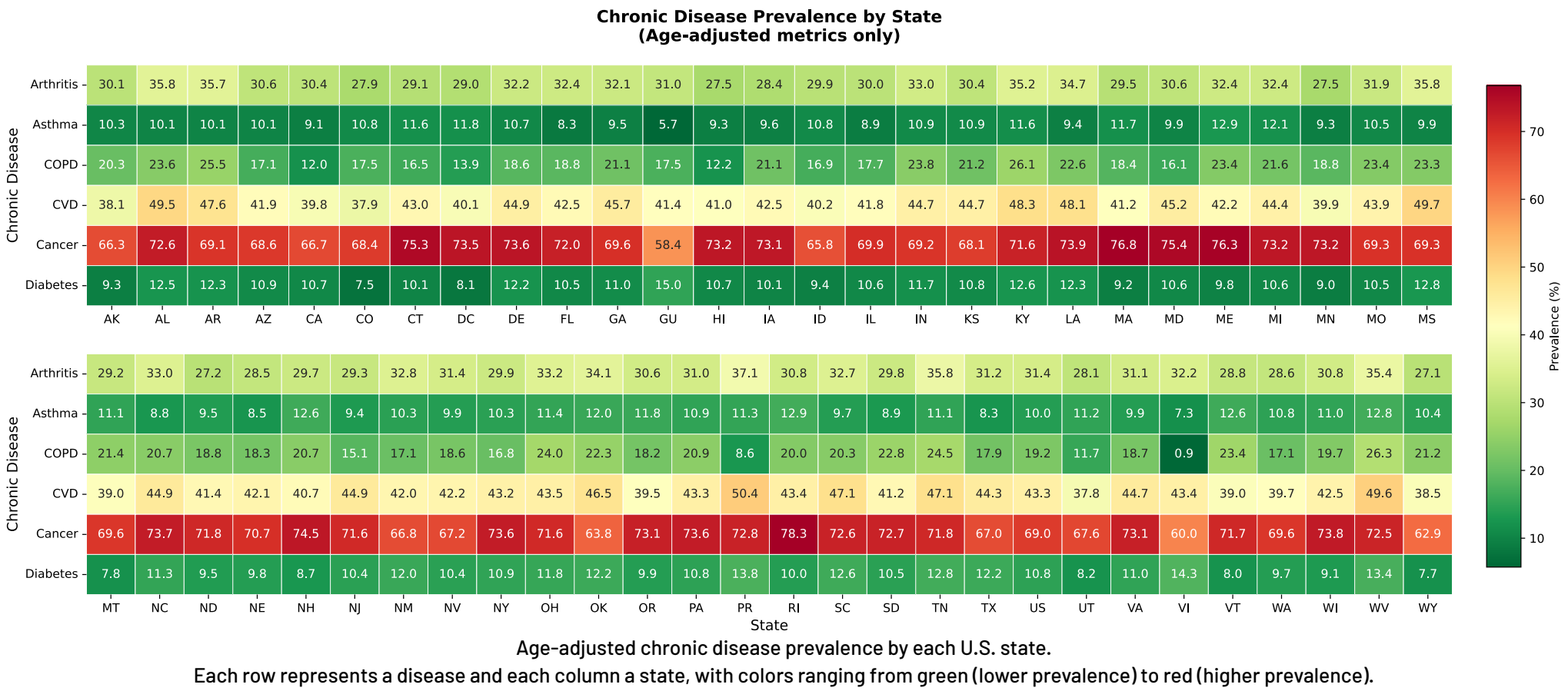
- Filter NaN values
- Stratification separation
- Metric normalization

VISUALIZATION

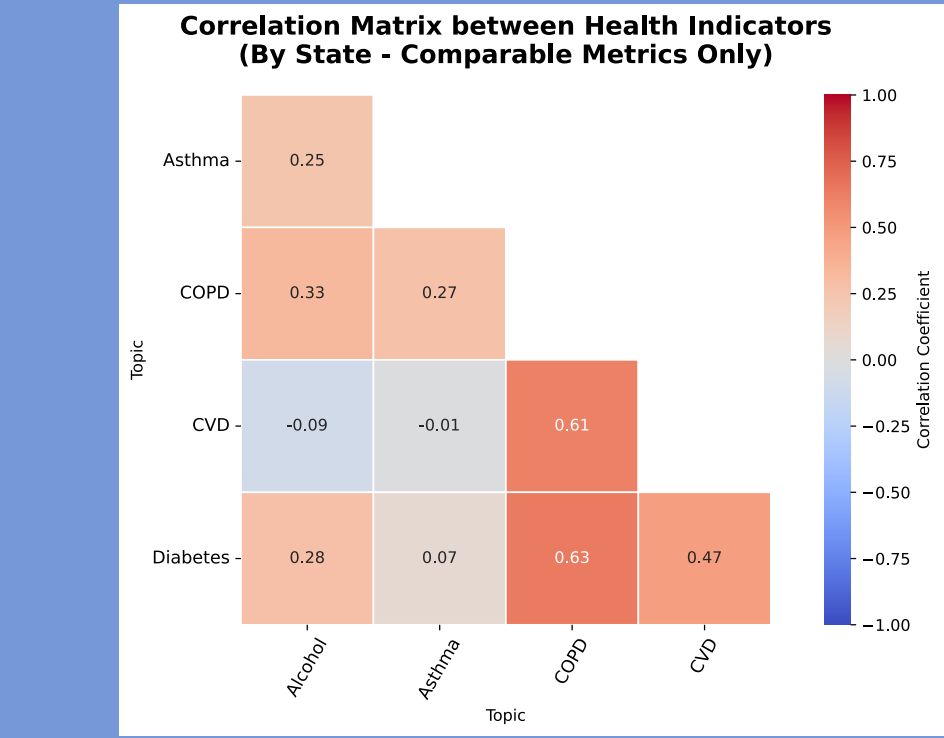
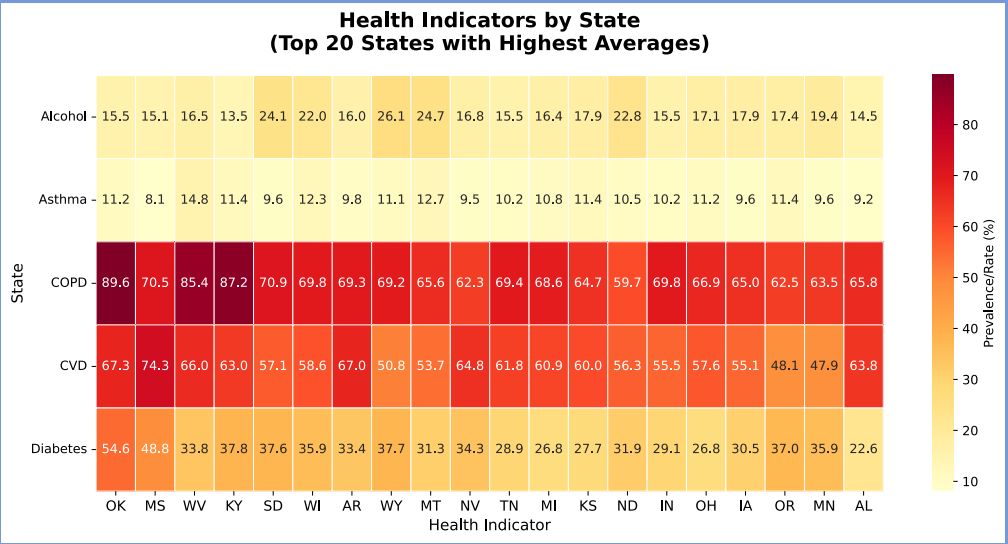
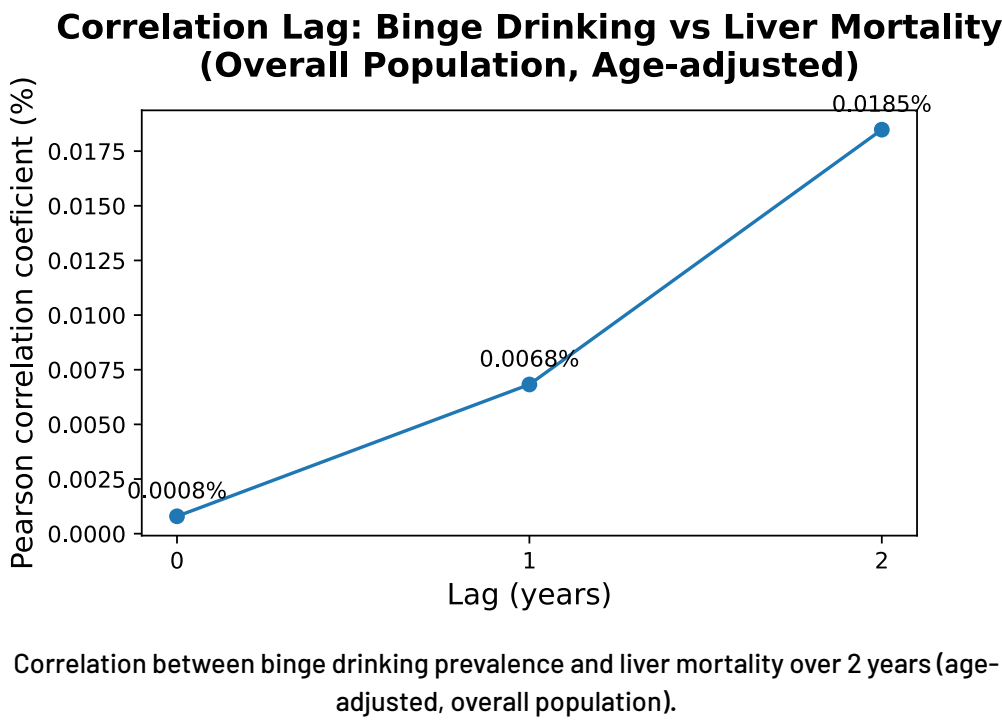
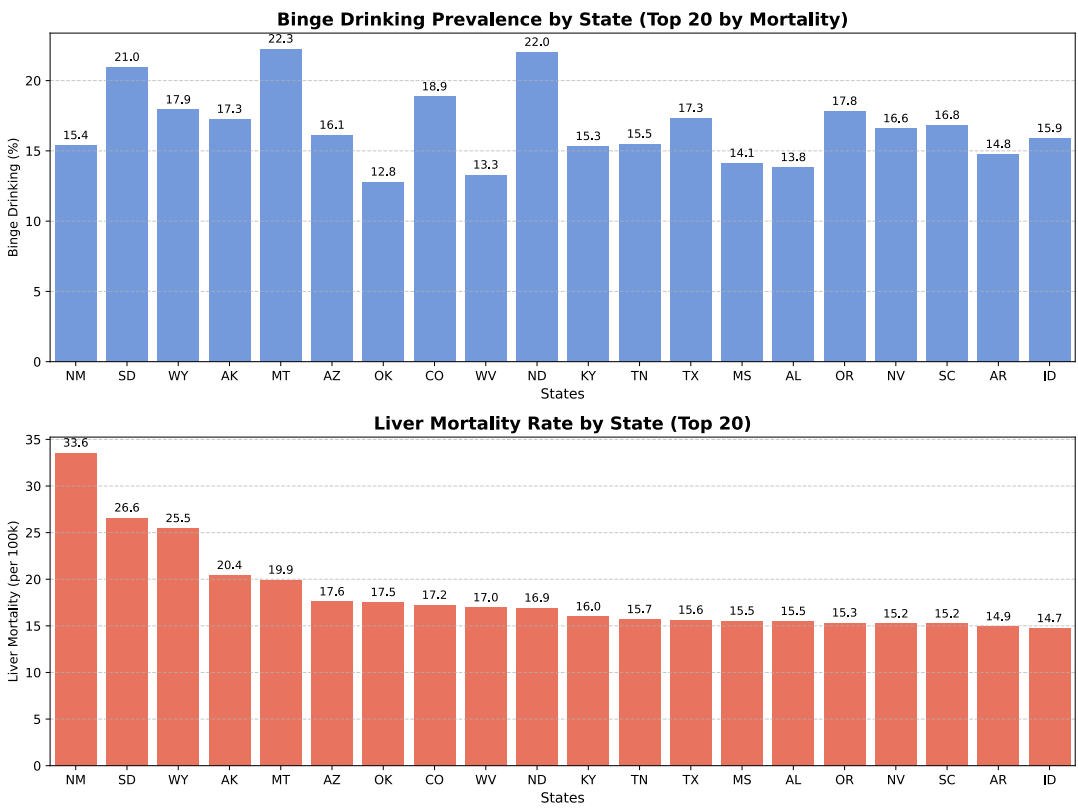
Visualizing data to help answer our questions

FEATURE EXTRACTION

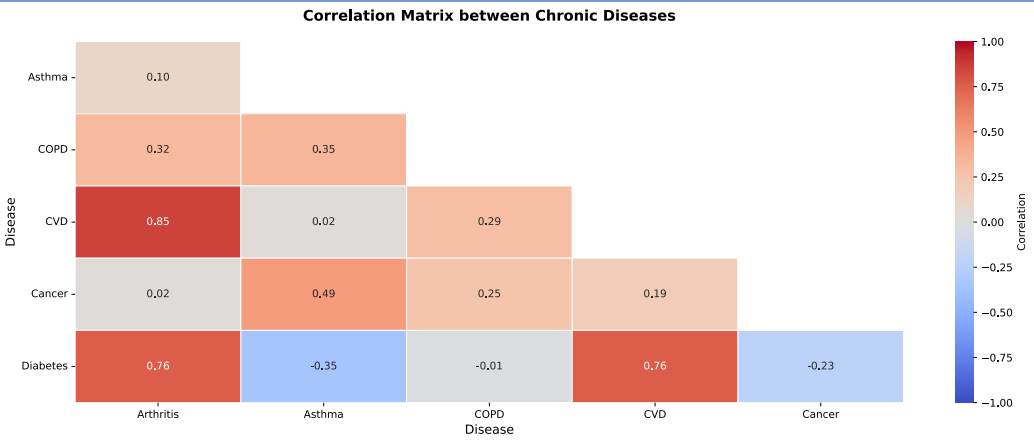
Extract columns to answer our questions



Relationship between binge drinking prevalence and liver mortality across U.S. states (age-adjusted). Each point represents a state, with binge drinking on the x-axis and liver mortality on the y-axis.



This correlation matrix shows the relationships between health indicators across states.



This correlation matrix illustrates the relationships between major chronic diseases across states.

Key findings and conclusion

- Binge drinking is directly related to liver mortality
- Strong correlations between metabolic and respiratory diseases
- States known for higher poverty and rurality are associated with worse chronic disease indicators
- Cancer has the highest prevalence across all states